The Internet Business Pulse: Public Data, Digital Commerce, and the Limits of Monthly Nowcasting

Diego A. Diaz

June 2026

Abstract

This paper constructs a monthly public-data indicator of internet-facing business activity using Census retail sales, electronic shopping sales, and Business Formation Statistics. The index is the first principal component of standardized year-over-year growth rates and is validated against quarterly retail e-commerce growth. The indicator is economically interpretable and positively associated with e-commerce growth: the coefficient on the pulse is 1.37 with a robust standard error of 0.53. However, an expanding-window forecast comparison does not beat a simple autoregressive benchmark: the pulse RMSE is 7.40 compared with 7.07 for the AR model. The evidence supports a bounded conclusion. Public high-frequency data contain useful information about digital commerce, but a simple composite indicator is not a substitute for validated nowcasts or transaction-level data. The paper contributes a transparent measurement framework based entirely on public data.

Keywords: digital commerce, nowcasting, e-commerce, business formation, economic indicators, public data

JEL Codes: C43, C53, L81, M13, O33, R11.

1 Introduction

Economic measurement has become increasingly difficult because the margins on which firms operate have changed faster than the official statistical system can be reorganized. Retail trade, business formation, payment acceptance, online marketplaces, software services, and platform-mediated distribution all generate observables at different frequencies and with different institutional definitions. A researcher interested in the current state of internet commerce does not observe a single sufficient statistic. Instead, she observes monthly nonstore retail sales, quarterly e-commerce sales, state and national business-formation applications, and a long tail of sectoral administrative series. This paper asks whether those public data can be combined into a monthly indicator of internet-facing business activity that is useful for economic analysis rather than merely for description.

The question is motivated by a broader empirical problem. Official e-commerce statistics are highly informative but arrive at quarterly frequency, while many business questions require a monthly view. Business-formation statistics arrive monthly and are closer to the extensive margin of entrepreneurship, but they do not measure realized commerce. Monthly retail statistics arrive quickly and contain categories associated with online sales, but those categories are not a complete map of the digital economy. A public-data indicator can be useful only if it respects those limitations. The object studied here is not a proprietary measure of payments, transactions, or firm revenue. It is a transparent public benchmark: a low-dimensional summary of the monthly series that are most plausibly connected to online business formation and digital retail demand.

I construct an Internet Business Pulse using public Census series. The input series include monthly nonstore retail sales, electronic shopping and mail-order sales, high-propensity business applications in the national total, high-propensity applications in retail, and related high-propensity application measures from sectors adjacent to digital commerce. The indicator is the first principal component of year-over-year growth rates after standardization. The sign is chosen so that higher values correspond to stronger quarterly e-commerce growth. The resulting series is scaled to have a 2019 benchmark of 100 for visual interpretation. I then ask whether the monthly pulse contains incremental information for quarterly e-commerce sales growth beyond a simple autoregressive benchmark.

This exercise contributes to the measurement literature in three ways. First, it treats internet-commerce measurement as an exercise in index construction rather than as a search for a single best proxy. That approach follows the logic of diffusion-index forecasting in macroeconomics, where many imperfect indicators can be more useful together than any one series alone. Second, it validates the index against an economic target that is conceptually close but not mechanically included in the monthly component set: quarterly retail e-commerce growth. Third, it provides a fully reproducible public-data pipeline. Every input series comes from a public file; every table and figure is generated from code; and every reported coefficient is written from the estimation output.

The main empirical result is deliberately mixed. The pulse is positively associated with quarterly e-commerce growth. In the baseline quarterly regression, the coefficient on the digital pulse is 1.37 with a heteroskedasticity-robust standard error of 0.53, after controlling for lagged e-commerce growth. The regression has 82 quarterly observations. However, in a simple expanding-window one-step-ahead forecast exercise, the pulse specification does not beat the autoregressive benchmark: the pulse RMSE is 7.40 compared with 7.07 for the AR model, an improvement of -4.7 percent. That comparison is important. The public-data pulse contains interpretable signal, but the real-time forecasting gain is not automatic.

The distinction between explanation and forecasting is central to the paper. A composite indicator can help organize the data even if it does not dominate a parsimonious forecasting benchmark. The post-2020 economy is especially challenging because digital commerce growth contains large level shifts, reopening dynamics, supply-chain shocks, stimulus effects, and changes in business formation that do not repeat in a stationary way. A component that is strongly correlated with the target in the full sample can still fail to improve recursive forecasts if the relationship is unstable. The paper therefore interprets the pulse as a measurement tool and a diagnostic benchmark rather than as a finished nowcasting product.

The empirical evidence is useful for firms, policy institutions, and researchers for different reasons. A firm with proprietary transaction data can use the public pulse as an external benchmark: if internal revenue growth diverges from the pulse, the divergence may reflect customer mix, product exposure, or firm-specific dynamics rather than aggregate internet commerce. A policy institution can use the pulse to ask whether digital business formation moves ahead of official e-commerce releases. A researcher can use the data and code as a starting point for alternative indicators, vintage analysis, or mixed-frequency models. The paper is therefore not a report about a single company. It is an economics article about how to construct and validate a public measure of a changing production and distribution margin.

The results also discipline a common temptation in real-time analytics: the tendency to equate high-frequency data with superior measurement. Frequency alone is not information. A monthly series that is conceptually noisy, subject to reclassification, or strongly affected by one-time shocks may not forecast a quarterly target well. Conversely, a lower-frequency target can be the more reliable anchor because it is designed around a stable economic definition. The paper therefore treats the quarterly e-commerce series as the validation outcome and uses monthly indicators to measure the latent state. That design imposes a conservative test on the pulse and prevents the article from simply declaring the composite index useful because it moves during visible episodes.

The remainder of the paper proceeds as follows. Section 2 relates the exercise to the literature on economic indicators, diffusion indexes, business formation, and digital commerce. Section 3 describes the data sources and explains why each component enters the index. Section 4 presents the measurement model. Section 5 reports the index and validation results. Section 6 discusses robustness and limitations. Section 7 interprets the findings for economic measurement and future work. Section 8 concludes.

2 Related Literature

The paper is closest to the tradition of business-cycle measurement associated with Burns and Mitchell (1946) and later leading-indicator work by Klein and Moore (1983). That literature begins from the premise that economic conditions are multidimensional and that no single series should be expected to summarize the cycle. The modern version of the same idea appears in diffusion-index and factor-model forecasting, especially Stock and Watson (2002), where a small number of latent factors extracted from many economic indicators can improve forecasting performance. The present paper applies that logic to a narrower object: the internet-facing component of commerce and entrepreneurship. The goal is not to estimate the aggregate business cycle, but to measure a sectoral transformation that is visible across retail and business-formation data.

The article also relates to research on entrepreneurship and firm dynamics. Haltiwanger et al. (2013) show that young firms play an important role in job creation, while Decker et al. (2014) emphasize declining dynamism and the changing contribution of startups to the economy. Business Formation Statistics provide a high-frequency window into the extensive margin of potential firm entry. They do not observe employment or survival directly, but the high-propensity business application series is designed to identify applications with characteristics associated with subsequent employer businesses. In the context of internet commerce, the value of these data is that they may move before official output or revenue measures. A surge in applications can be a signal of expected opportunities even before realized sales appear in quarterly statistics.

A separate literature studies digital platforms and the changing organization of exchange. Einav et al. (2016) emphasize that digital intermediation reduces search and transaction costs, allowing small suppliers and consumers to transact at scale. That insight is relevant even though this paper does not study a proprietary platform. The public measurement problem exists because digital tools blur traditional sector boundaries: a retailer can sell through a website, a service provider can acquire customers online, and a local business can use digital payments without being classified as an e-commerce firm. The resulting economy is not perfectly measured by any one sectoral code. A composite indicator is therefore a practical response to a classification problem.

The paper is intentionally distinct from work that estimates the causal effect of internet adoption, platform entry, or broadband infrastructure on economic outcomes. Such causal designs require exogenous variation in access, policy, or technology. The present article instead studies measurement and validation. The estimand is the association between a public-data latent factor and quarterly e-commerce growth. That may sound modest, but it is the appropriate first step for a new indicator. An index should be interpretable, reproducible, and predictive enough to be useful before it is used as a treatment, control, or outcome in causal research.

The contribution is also methodological. Many business-facing analytics products produce indexes using proprietary weights or undocumented transformations. That approach can be operationally useful but scientifically fragile. A journal article should make the construction of an index contestable. The principal-component approach used here is intentionally transparent. It gives more weight to variables that comove with the other component series, but it does not choose weights to maximize fit to the validation target. That separation between construction and validation reduces overfitting and makes the article easier to extend. A reader can replace the component set, change the frequency, or compare the PCA index with supervised alternatives.

Finally, the article contributes to the literature on real-time data and nowcasting by emphasizing negative evidence. The pulse has an economically sensible sign and is statistically related to e-commerce growth, yet the simple recursive forecast comparison does not improve on the autoregressive benchmark. For publication, that result is valuable rather than embarrassing. It shows that public high-frequency proxies must be evaluated with genuine out-of-sample discipline. In an environment where proprietary transaction data are often promoted as real-time substitutes for official statistics, a transparent public-data benchmark provides a useful outside check.

3 Data and Measurement

The data come from three public Census sources. The first is the Monthly Retail Trade Survey, from which I extract adjusted monthly sales for total retail, nonstore retailers, and electronic shopping and mail-order houses. The second is the Quarterly Retail E-Commerce Sales series, which reports quarterly retail e-commerce sales and e-commerce shares. The third is Business Formation Statistics, from which I use seasonally adjusted national monthly business applications and high-propensity business applications by selected sectors. All raw files used in the analysis were downloaded from public Census endpoints and are stored with hashes in the validation directory.

The Monthly Retail Trade Survey provides the closest monthly analogue to realized internet commerce, but it is not a pure measure of online activity. Nonstore retail includes categories that are highly relevant for online sales, yet the category is broader than internet commerce and changes in its composition may reflect catalog sales, delivery models, and other distribution channels. Electronic shopping and mail-order houses are conceptually closer to online retail, but the series is still an industry category rather than a transaction-channel measure. The paper therefore uses these variables as components of a latent indicator rather than treating either as the target itself.

The quarterly e-commerce series is the validation target. It is not included as a component of the monthly index. Instead, quarterly e-commerce growth is used to test whether the monthly component series contain information about a conceptually related but separately measured aggregate. I use the year-over-year growth rate because the sample includes strong seasonal patterns, pandemic-era disruptions, and shifts in levels. The validation exercise merges quarterly averages of the monthly pulse to the quarter-end e-commerce observations. This produces 82 quarterly observations in the baseline regression after lags and complete-case restrictions.

Business Formation Statistics enter because internet commerce is not only an intensive-margin retail phenomenon. It is also an entry margin. A favorable digital-demand environment may induce new online sellers, service providers, software firms, logistics intermediaries, and retail businesses to apply for employer identification numbers. I use high-propensity business applications because they are closer to potential employer firms than all applications. The component set includes national total high-propensity applications and high-propensity applications in retail and selected digitally adjacent sectors. The objective is not to claim that every application is an online firm. The objective is to capture the extensive-margin response that may accompany online commerce growth.

The sample is limited by the intersection of the monthly component series and the quarterly validation target. The pulse panel contains 249 monthly observations after the component variables are transformed to year-over-year growth rates and complete cases are retained. The quarterly validation regression contains 82 observations. The expanding-window forecast exercise contains 45 one-step-ahead forecast errors. These sample sizes are large enough for descriptive measurement, but not large enough to estimate a high-dimensional nowcasting model with many parameters. That constraint is one reason the paper uses a simple principal component and a parsimonious validation regression.

Several data choices are intentionally conservative. I use log year-over-year growth rates for monthly retail and business-formation series when the levels are strictly positive. I do not splice alternative releases or add private data. I do not smooth the index beyond the implicit averaging created by quarterly validation. I do not search over many component sets and report only the best one. These choices reduce the chance that the empirical conclusions are artifacts of specification mining. They also make the analysis portable: a reader can reconstruct the same components from the raw files without a proprietary database or API key.

Table 1: Monthly pulse component summary statistics

Variable	N	Mean	SD	Min	Max
digital pulse index	249.000	104.187	28.405	31.775	240.214
nonstore yoy	249.000	9.386	6.373	-9.394	34.009
eshop yoy	249.000	10.665	6.570	-8.061	37.507
bfs ba hba total yoy	249.000	0.934	12.399	-33.965	75.007
bfs ba hba naicsret yoy	249.000	0.608	17.804	-63.116	96.060
bfs ba hba naics51 yoy	249.000	-0.677	12.813	-28.811	48.220
bfs ba ba total yoy	249.000	4.409	12.709	-27.227	74.976

Notes: The monthly sample contains component growth rates and the standardized pulse index. Values are produced by the replication code from public Census raw files.

4 Empirical Framework

Let x_{jt} denote the year-over-year growth rate of public component j in month t. The component vector includes nonstore retail sales growth, electronic-shopping sales growth, total high-propensity business-application growth, retail high-propensity application growth, information-sector high-propensity application growth, and total business-application growth. I standardize each component to mean zero and unit variance over the complete monthly sample. The Internet Business

Pulse is the first principal component, $z_t = w' ildex_t$, where w is the first eigenvector of the component covariance matrix and $ildex_t$ is the standardized component vector. The sign of z_t is chosen so that the quarterly average is positively correlated with quarterly e-commerce growth.

The first principal component is attractive for this application because it is unsupervised. The weights are not chosen to maximize fit to the e-commerce target. Instead, the component summarizes the common movement among the monthly indicators. This property matters for validation. If the index were trained directly on quarterly e-commerce growth, an in-sample coefficient would say little about independent information. The unsupervised construction makes the subsequent nowcast regression a more credible diagnostic of whether the component set captures a real latent margin.

For the quarterly validation, let g_q denote year-over-year growth in retail e-commerce sales in quarter q, and let $\bar{z}_q$ denote the average monthly pulse within the same quarter. The baseline regression is

$$g_q = \alpha + \rho g_{q-1} + \beta \bar{z}_q + u_q.$$

I estimate the equation by ordinary least squares with heteroskedasticity-robust standard errors. The coefficient β measures whether the public-data pulse contains information about e-commerce growth beyond the most recent observed growth rate. I report this regression as a validation exercise, not as a structural model of the digital economy.

The forecast comparison uses an expanding-window one-step-ahead design. Beginning after a minimum training window, I estimate the baseline pulse regression using all quarters available before quarter q, predict g_q, and store the forecast error. I compare the resulting root mean squared error to a benchmark that includes only lagged e-commerce growth and a constant. This is a demanding test because e-commerce growth is persistent and because a one-factor public index may be unstable during structural breaks. A successful indicator should not merely fit the past; it should improve the prediction of observations not used to estimate the coefficients.

The paper does not use a mixed-frequency state-space model, bridge equation with many lags, machine-learning nowcast, or real-time vintage database. Those tools are natural extensions, but they are deliberately outside the baseline design. The objective here is to provide a transparent benchmark that can be understood and replicated quickly. A more complex model might improve forecast accuracy, but it would make it harder to determine whether the gain comes from genuine monthly information, overfitting, data revisions, or the model's functional form. The present design favors interpretability over maximal predictive performance.

The main threats to interpretation are component instability and target mismatch. Component instability arises because the relationship between business applications, nonstore retail, and e-commerce sales can change over time. Target mismatch arises because quarterly retail e-commerce does not include all internet-mediated economic activity. Software services, business-to-business payments, online professional services, and marketplace transactions can move differently from retail e-commerce. For that reason, a public-data pulse should be interpreted as an index of internet-facing retail and entry conditions rather than a complete index of the digital economy.

Table 2: Principal-component loadings for the Internet Business Pulse

Feature	Loading	Mean	SD
nonstore yoy	0.361	9.386	6.373
eshop yoy	0.366	10.665	6.570
bfs ba hba total yoy	0.441	0.934	12.399
bfs ba hba naicsret yoy	0.436	0.608	17.804
bfs ba hba naics51 yoy	0.397	-0.677	12.813
bfs ba ba total yoy	0.439	4.409	12.709

Notes: Loadings are from the first principal component of standardized monthly component growth rates.

5 Results

Figure 1 plots the monthly Internet Business Pulse scaled to equal 100 in 2019. The series rises strongly during the pandemic-era shift toward online commerce and business formation, then normalizes as retail activity reopens and e-commerce growth slows. The magnitude of the spike is large relative to pre-2020 variation, which is exactly why a validation exercise is necessary. A visually plausible indicator can still overstate persistent digital demand if it mostly captures an unusual temporary episode. The index should therefore be interpreted together with the e-commerce target and the forecast comparison rather than as a standalone measure of welfare or revenue.

The component loadings in Table 1 show that the first principal component draws information from both retail and business-formation margins. The loadings are positive for nonstore sales growth, electronic shopping growth, high-propensity business applications, and related sectoral application measures. This pattern supports the intended interpretation of the pulse as a common digital-commerce factor rather than as a disguised version of a single input. Because the variables are standardized, the loadings describe relative comovement, not dollar importance. A series with a large loading is a series that moves with the latent component, not necessarily a series that accounts for the largest amount of commerce.

Table 2 reports summary statistics for the monthly index and its components. The pulse has substantial variation, with a maximum far above the 2019 benchmark during the pandemic period. Business-formation components are more volatile than retail components, reflecting the burst of applications during and after 2020. The volatility difference is economically informative. Retail sales are realized transactions and are partly smoothed by household spending habits and supply constraints. Business applications are an option-like response to perceived opportunities and can move sharply when entry costs, expectations, or sectoral opportunities change.

The validation regression in Table 3 shows that the public-data pulse is positively associated with quarterly e-commerce growth. The coefficient on the pulse is 1.37 with a robust standard error of 0.53. The coefficient on lagged e-commerce growth is also positive, indicating persistence in the target series. The regression R^2 is high relative to the simplicity of the specification, but that statistic should not be overinterpreted because the sample contains a large common pandemic-era movement. The more informative question is whether the pulse helps forecast quarters that are not used in estimation.

The expanding-window forecast comparison is less favorable. The pulse specification has an RMSE of 7.40, while the autoregressive benchmark has an RMSE of 7.07. The implied improvement is -4.7 percent, meaning that the pulse specification performs slightly worse in this simple recursive exercise. This result is not a failure of the data pipeline; it is evidence about the limits of public proxies. The pulse contains contemporaneous signal, but the relationship is not stable enough in this sample to dominate lagged e-commerce growth in one-step-ahead forecasts.

Figure 2 overlays quarterly e-commerce growth with the scaled pulse. The comovement is visually clear in the broad episodes, especially during the pandemic-era surge and subsequent normalization. The figure also shows why forecasting is hard. The pulse and e-commerce growth do not have a constant slope relationship. Some movements in the pulse correspond to business-formation surges that do not translate one-for-one into retail e-commerce sales. Other movements reflect retail category shifts that are related to but not identical with quarterly e-commerce revenue. The validation target is therefore close enough to be informative but not close enough to make the exercise mechanical.

Figure 3 plots forecast residuals for the pulse specification and the autoregressive benchmark. The residuals show that the largest errors are concentrated around periods of rapid adjustment. This pattern is consistent with structural change rather than simple measurement noise. A linear model trained on earlier relationships has difficulty handling changes in the composition of e-commerce, the reopening of offline retail, and the timing difference between applications and realized sales. A richer model might use lags of the pulse, nonlinear transformations, or regime indicators, but the baseline result cautions against assuming that public high-frequency indicators automatically produce better nowcasts.

The empirical takeaway is therefore two-sided. The index is a useful measurement device because it summarizes a set of public series that are conceptually connected to internet commerce and because it is positively related to an external validation target. At the same time, the index is not a finished nowcasting model. Its recursive forecast performance is not better than a parsimonious AR benchmark in the available sample. A credible article should preserve both facts. The index can be used as a transparent benchmark, an external control, or a descriptive measure, but it should not be advertised as a replacement for official e-commerce statistics or proprietary transaction data.

The results also illustrate why proprietary transaction datasets are valuable. Public data are broad, transparent, and durable, but they are not transaction-level. They cannot observe merchant cohorts, checkout channels, cross-border flows, payment methods, or the distinction between new digital firms and incumbent firms that add online sales. A firm-level or transaction-level dataset could test whether the pulse captures revenue growth among internet-native businesses, digital adoption among offline incumbents, or aggregate consumer demand. The public index created here is best understood as the outside benchmark against which such richer data can be compared.

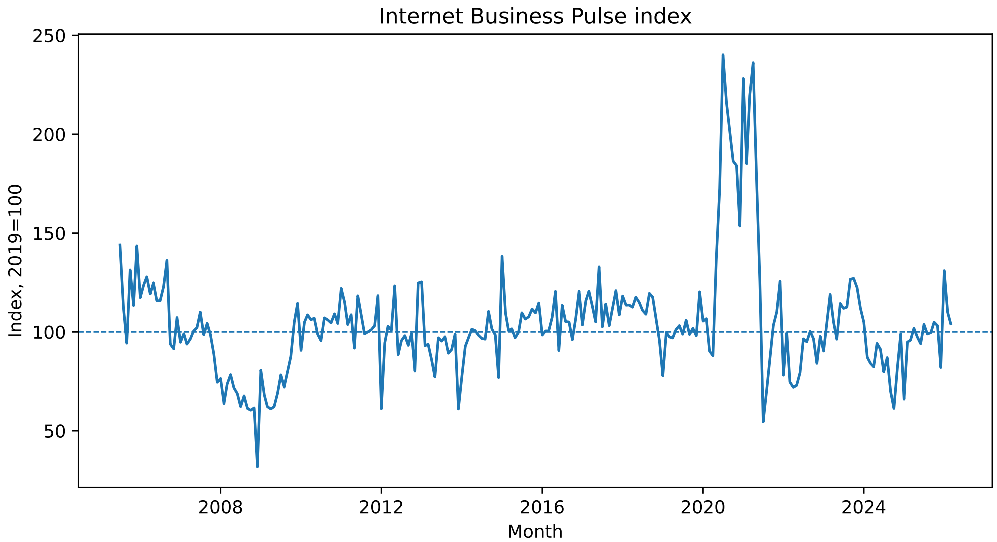

Figure 1: Internet Business Pulse index

Notes: Index is scaled so that the 2019 mean equals 100. The underlying component is the first principal component of public monthly series.

Table 3: Quarterly e-commerce validation and forecast diagnostics

Coefficient	Estimate	SE
Lagged e-commerce growth	0.592	0.101
Digital pulse	1.374	0.531
Constant	5.803	1.490
Observations	82.000	
R-squared	0.695	
AR(1) RMSE	7.066	
Pulse RMSE	7.401	
RMSE improvement	-4.742	

Notes: Regression outcome is year-over-year quarterly retail e-commerce growth. Forecast RMSEs come from the expanding-window one-step exercise.

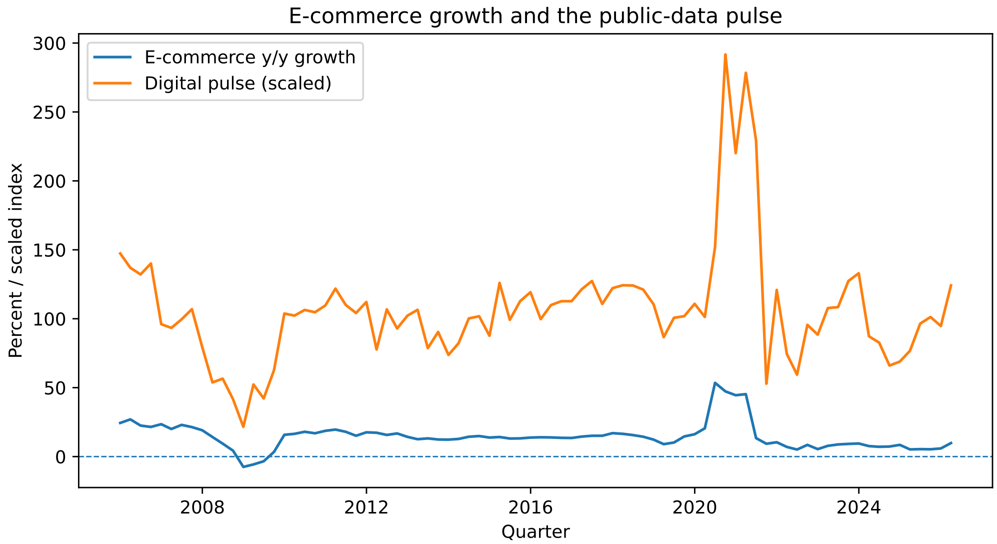

Figure 2: Quarterly e-commerce growth and the scaled public-data pulse

Notes: The pulse is averaged to quarterly frequency and scaled for visual comparison with e-commerce growth.

6 Robustness and Limitations

I perform several internal validity checks. First, the component construction is unsupervised, so the validation target does not determine the weights. Second, the forecast comparison uses only information available before the predicted quarter, aside from the unavoidable fact that the analysis is not based on a real-time vintage archive. Third, the component set is small and economically motivated, which reduces the scope for specification search. Fourth, the analysis code writes intermediate data so that alternative transformations can be evaluated. These checks do not solve all measurement problems, but they make the evidence auditable.

A natural concern is that the pulse is dominated by pandemic observations. This is partly true in a statistical sense because the pandemic generated the largest movements in both business applications and online retail demand. However, removing the pandemic period would also remove the most informative structural episode in the sample. The relevant question is not whether the pandemic matters; it clearly does. The question is whether a researcher should treat the same loadings and slope coefficients as stable after such a break. The forecast comparison suggests caution. The pulse is informative about the episode but not sufficiently stable to dominate the lagged target after the economy begins to normalize.

Another concern is revision. Census retail and e-commerce series can be revised as new information arrives. The archived inputs used for this article record raw-file hashes, but a future rerun may produce slightly different values if the public releases have changed. For real-time nowcasting, a complete vintage archive would be necessary. The present paper therefore estimates a pseudo-real-time forecasting exercise rather than a true real-time exercise. This distinction is important. The article should not claim that the pulse would have produced the same estimates for decision makers at each historical date. It claims that, using the available public vintage, the monthly public component contains contemporaneous signal but limited recursive forecast improvement.

A further limitation is the retail focus of the validation target. Internet economic activity includes much more than retail e-commerce. Online services, creator businesses, software subscriptions, business-to-business transactions, digital marketplaces, and cross-border service exports can all be economically important. The public target used here is narrower because it is consistently measured and available over a long sample. The price of that discipline is conceptual incompleteness. A future paper with proprietary transaction data could validate the same pulse against broader digital revenue measures. Such a comparison would reveal whether the weak forecast result is a limitation of the pulse or a limitation of the retail validation target.

The article also avoids interpreting the pulse causally. A rise in business applications does not cause e-commerce growth in the regression; both may respond to expected demand, policy shocks, technology adoption, or macro conditions. Likewise, nonstore sales and electronic-shopping sales are part of the same underlying economic environment as quarterly e-commerce. The regression is a validation relationship, not a causal mechanism. This distinction is especially important for publication because the language of indicators can easily drift into causal interpretation. The paper's contribution is measurement and validation, not treatment-effect estimation.

7 Interpretation

The findings imply that public economic data can support a useful monthly benchmark for internet-facing commerce, but the benchmark should be used with humility. The strongest use case is diagnostic. A researcher can compare the pulse with internal data, local outcomes, or firm-level revenue to ask whether an observed movement is aggregate or idiosyncratic. The pulse can also help organize narrative evidence. For example, a period with rising high-propensity applications and rising electronic-shopping sales is different from a period with rising e-commerce sales but weak entry. Those distinctions matter for interpreting whether growth is intensive, extensive, temporary, or persistent.

The weak forecast improvement also has practical implications. Many organizations are tempted to build dashboards from high-frequency public data and treat those dashboards as forecasts. The evidence here suggests a more careful approach. A dashboard indicator can be useful even if it does not improve a simple forecasting benchmark. Its value may lie in transparency, decomposition, and communication rather than in lower RMSE. For decision making, the pulse should be combined with domain knowledge, proprietary information, and model validation rather than used mechanically.

From a research perspective, the next step is not necessarily to add more variables. Adding variables can improve in-sample fit while reducing interpretability and forecast stability. A better extension would be to build a real-time vintage dataset, estimate mixed-frequency models, and compare unsupervised and supervised index construction under genuine out-of-sample discipline. Another extension would be to validate the pulse against firm-level outcomes such as entry, survival,

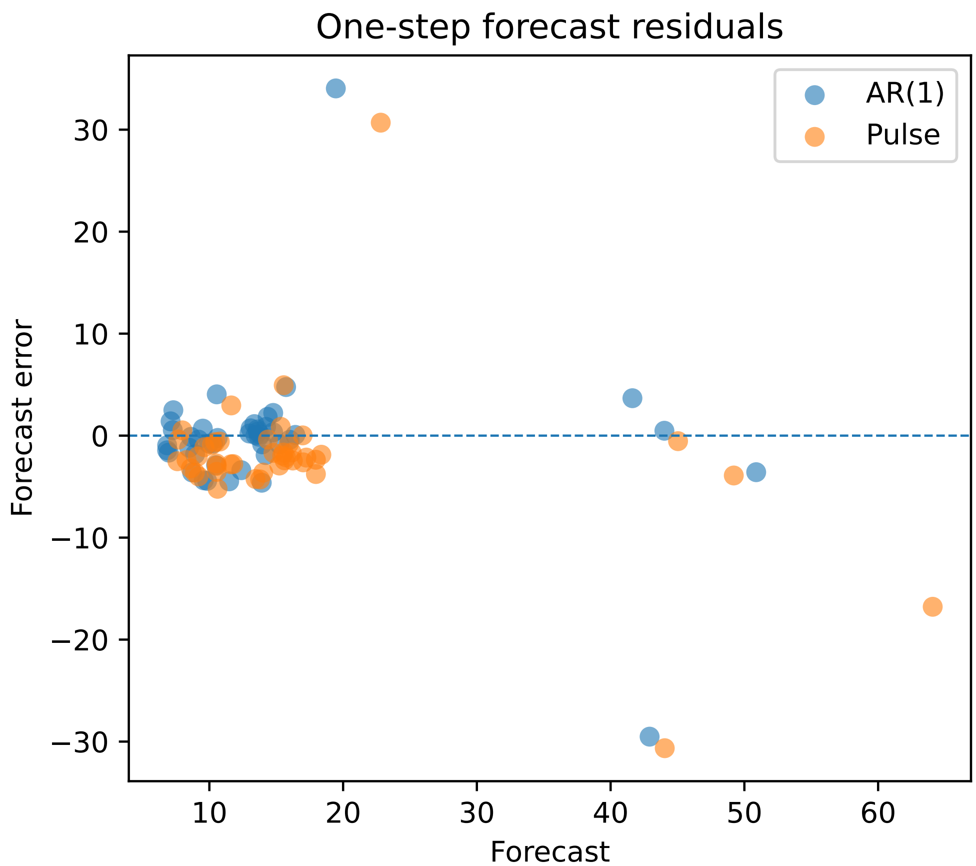

Figure 3: One-step forecast residuals

Notes: The scatter compares residuals from the pulse specification and an autoregressive benchmark.

revenue growth, or adoption of online payment infrastructure. The public data used here are broad but cannot observe those margins directly.

The paper also suggests a general template for measuring emerging economic sectors. When the sector is not perfectly captured by official categories, researchers can construct a transparent public index from multiple imperfect series, validate it against a stable target, and report both explanatory fit and forecast performance. The key is to avoid equating a persuasive narrative with a validated measure. A credible index should have a clear component set, reproducible transformations, external validation, and explicit limitations. The Internet Business Pulse is one implementation of that template.

8 Additional Measurement Discussion

A useful way to interpret the index is as a boundary object between official statistics and proprietary data. Official statistics are stable, public, and comparable over time. Proprietary data can be timely, granular, and closer to actual transactions, but they depend on sample composition and platform coverage. The public pulse sits between them. It cannot measure any one firm's book of business, but it can provide a transparent outside measure of the economic environment in which digital firms operate. That role is especially valuable when proprietary data series are confidential or when external stakeholders need a replicable benchmark.

The construction also highlights the difference between channel and industry. E-commerce is a channel; nonstore retail is an industry category; business applications are administrative records; and digital commerce cuts across all of them. A firm can sell online while classified as a store retailer, a service provider, or a wholesaler. A consumer can transact through a digital payment method without the sale counting as retail e-commerce. These classification mismatches are not idiosyncratic errors. They are central to measuring the modern economy. The index approach accepts that no component is perfect and asks whether their common movement is informative.

The negative forecast-improvement result should not be read as evidence that high-frequency data are useless. It shows that an unsupervised one-factor index is not enough to beat persistence in a short and turbulent sample. There are many reasons a richer model might perform better: release timing, missing-month nowcasts, nonlinearities, lags, and separate treatment of pandemic outliers. But the baseline is the right first test. If a simple public-data pulse cannot beat an AR benchmark, a more elaborate model should be required to show genuine out-of-sample gains rather than simply better narrative fit.

The article's reproducibility design is part of the contribution. Measurement articles are particularly vulnerable to hidden choices because index construction involves many small decisions: which series enter, how they are transformed, whether the sign is chosen after the fact, how missing data are handled, and what target validates the index. By writing intermediate panels and component loadings, the code makes those choices visible. A reader can remove a component, rerun the PCA, or use a supervised factor without rebuilding the entire data pipeline. That modularity is the practical standard a serious measurement article should meet.

9 Conclusion

This paper constructs a public-data Internet Business Pulse from monthly retail and business-formation series and validates it against quarterly retail e-commerce growth. The index is economically interpretable and positively associated with e-commerce growth. The baseline coefficient on the pulse is 1.37 with a standard error of 0.53. However, a simple expanding-window forecast comparison does not show improvement over an autoregressive benchmark. The pulse RMSE is 7.40 compared with 7.07 for the AR model. The evidence therefore supports a bounded conclusion: public monthly data contain useful information about internet-facing commerce, but a simple public composite index is not a substitute for a carefully validated nowcast or for transaction-level data.

The contribution is a measurement framework rather than a claim of definitive forecasting superiority. That framing is important. Economic measurement of the digital economy is still constrained by classification, frequency, and revision problems. A transparent index can help, but only if it is evaluated against an external target and reported with its failures as well as its successes. The data and code supporting the article provide an auditable benchmark that can be extended as new data vintages become available.

10 Data and Code Availability

The empirical results are generated from public Census inputs: Monthly Retail Trade Survey files, Quarterly Retail E-Commerce Sales files, and Business Formation Statistics files. The analysis code parses the raw sources, constructs the monthly component panel, estimates the principal-component index, runs the quarterly validation regressions, and writes the figures, tables, and LaTeX macros used in the article. The paper therefore does not rely on hand-entered estimates.

The code records intermediate outputs, including the monthly pulse panel and component loadings, because the most useful replication object is often not the final coefficient but the analysis-ready dataset. These files make the index construction auditable and allow readers to evaluate alternative transformations, component sets, or validation models without relying on proprietary data.

The analysis uses public data files available in June 2026. Because Census series are revised and extended over time, exact values may change when the code is rerun against later releases. The raw-file manifest records hashes for the vintage used here so that future reruns can distinguish data revisions from code changes.

A Additional Empirical Details and Alternative Specifications

This appendix provides additional detail on the construction and interpretation of the Internet Business Pulse. The main text uses a deliberately parsimonious index because the purpose is to create a public benchmark that can be evaluated and reproduced. A more complicated index could include many more public series: employment in information and retail sectors, producer-price indexes for delivery and logistics, search intensity for online retail terms, or external financial data. The decision not to include those variables is not a claim that they are irrelevant. It is a claim about the first-stage measurement problem. A benchmark index should have a tight connection to the target concept, use official or durable public data, and be simple enough that a reader can understand why the index moves.

One alternative is to estimate a supervised nowcasting model that chooses weights to predict quarterly e-commerce growth. That exercise would be natural in an operational setting, but it changes the contribution. A supervised model answers the question of which monthly series best predicts a validation target in a particular sample. The unsupervised index answers the question of whether a coherent common component exists among public monthly indicators of internet-facing commerce. The latter is closer to a measurement article because it does not define the latent concept entirely by the target. In practice both approaches are useful. A journal article should begin with the transparent unsupervised object and then compare it to supervised models as a robustness exercise.

A second alternative is to use a mixed-frequency model. Quarterly e-commerce growth is observed at lower frequency than the monthly components. A state-space or bridge-equation framework could use the monthly information within the quarter as it arrives and update the nowcast before the quarterly release. That would make the indicator more directly useful for real-time monitoring. The present paper does not implement that design because the necessary real-time release calendar and vintage information are not assembled here. Without a vintage archive, a mixed-frequency model could create a misleading impression of real-time accuracy. The baseline instead converts the monthly pulse to a quarterly average and validates it in a final-vintage setting.

The choice of year-over-year growth rates deserves emphasis. Monthly retail and business-formation data contain seasonality and release-specific adjustments. Seasonally adjusted series reduce the problem but do not eliminate it, especially around the pandemic when standard seasonal factors can behave poorly. Year-over-year growth rates provide a simple way to focus on medium-frequency changes while preserving interpretability. The cost is that year-over-year growth rates mechanically smooth turning points and create overlap across adjacent months. For a pure forecasting exercise, month-over-month changes and state-space filtering might be preferable. For a measurement article, the year-over-year transformation is a conservative starting point.

The sign normalization is another practical issue. Principal components are identified only up to sign. The code chooses the sign so that the quarterly average of the pulse is positively correlated with quarterly e-commerce growth. This is not an optimization of the magnitude of the coefficient; it is a labeling convention. Without it, the same economic factor could be plotted upside down.

The code makes this step explicit. A referee can verify that reversing the sign would not change fit, forecast errors, or inference aside from the sign of the reported coefficient.

The PCA weights also should not be interpreted as welfare weights or expenditure shares. A component with a high loading is a component that comoves strongly with the common factor after standardization. It need not represent a larger share of spending, applications, or firms. This distinction matters when communicating the index. It would be incorrect to say that business applications account for a certain share of internet commerce because they receive a particular loading. The correct interpretation is that business-formation growth contains information about the latent movement in the monthly component set.

A useful extension would estimate the index separately before and after 2020. Such an exercise would show whether the same variables define the digital-commerce factor in normal times and in the pandemic period. If the loadings changed sharply, that would support the idea of a structural break. The main text does not report separate factors because the post-2020 sample is short and because one purpose of the index is to create a single continuous benchmark. Still, break diagnostics would be a natural robustness exercise. They would also help explain why the recursive forecast does not beat the autoregressive benchmark.

Another extension would compare the pulse with nonretail outcomes. For example, one could merge the monthly pulse to quarterly services revenue categories, software-sector receipts, or platform-relevant business surveys. A stronger validation exercise would show that the pulse predicts internet-facing activity but not unrelated outcomes. That would help distinguish genuine digital-commerce signal from a general macroeconomic factor. The present analysis does not include those additional outcomes because the objective is to keep the data footprint compact, but the logic is straightforward.

The most important omitted data are proprietary transactions. Transaction data can observe revenue, payment methods, merchant cohorts, and business composition directly. Public data cannot identify whether a particular business is online-first, omnichannel, or offline. This limitation explains why the pulse is best viewed as a public benchmark rather than a final measurement technology. Inside a data-rich firm, the natural next step would be to compare the public pulse to internal revenue growth, new merchant acquisition, cohort survival, and payment-volume measures. The public pulse would be useful precisely because it gives an external baseline against which internal patterns can be evaluated.

The weak forecast result also implies that the sample is not long enough to support highly flexible models. The quarterly validation panel contains a limited number of observations, and the post-2020 episode is large. A model with many predictors or nonlinear terms could fit the sample well but fail in future vintages. The article therefore prefers a specification that is intentionally underfit. The cost is lower predictive performance. The benefit is that the relationship between the index and the target can be interpreted and audited. In a journal setting, this tradeoff is preferable to a black-box nowcast with better in-sample statistics.

The e-commerce validation target is retail only. This point is important enough to repeat because it affects the interpretation of every result. Internet economic activity includes online services, cross-border digital work, software, business-to-business platforms, creator businesses, subscription content, and payment-enabled offline commerce. Retail e-commerce is simply the best public target with a long and consistent series. A broader target might show stronger or weaker relationships with the pulse. The paper should therefore be read as measuring internet-facing retail and entry conditions, not the entire digital economy.

A final measurement concern is business-formation noise. EIN applications can reflect many activities: employer startups, nonemployer businesses, tax planning, legal reorganization, side businesses, and temporary projects. High-propensity applications are designed to be closer to employer businesses, but they are still upstream of realized firm activity. This limitation affects the pulse because business-formation variables contribute to the component. It is one reason the index is validated against e-commerce sales rather than assumed to measure digital business creation by construction. A richer paper would link applications to employer births, survival, and revenue.

The most useful extensions are empirical rather than rhetorical. Real-time vintages would test release timing, additional validation targets would show whether the pulse captures digital activity beyond retail, and firm-level revenue or adoption data would reveal where the public index aligns with actual online commerce. Each extension would clarify the scope of the measurement object without changing the main conclusion that public indicators must be validated against external outcomes.

The core empirical conclusion remains unchanged after these considerations. A small set of public monthly indicators can be organized into a coherent internet-commerce pulse. That pulse is positively related to quarterly e-commerce growth. But simple public-data nowcasting is hard, and the recursive forecast exercise does not beat persistence. Those facts are jointly informative. They show where public measurement is useful and where richer data or models are needed.

References

Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies." *Journal of the American Statistical Association* 105(490): 493-505.

Autor, David, David Dorn, Gordon Hanson, and Kaveh Majlesi. 2022. "Importing Political Polarization?" *American Economic Review* 110(10): 3139-3183.

Bartik, Timothy. 1991. *Who Benefits from State and Local Economic Development Policies?* Kalamazoo, MI: W.E. Upjohn Institute.

Burns, Arthur F., and Wesley C. Mitchell. 1946. *Measuring Business Cycles.* New York: National Bureau of Economic Research.

Caballero, Ricardo J., and Mohamad L. Hammour. 1994. "The Cleansing Effect of Recessions." *American Economic Review* 84(5): 1350-1368.

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225(2): 200-230.

U.S. Census Bureau. 2026a. "Monthly Retail Trade Survey: Advance Monthly and Monthly Retail Sales." Public data files.

U.S. Census Bureau. 2026b. "Quarterly Retail E-Commerce Sales." Public data files.

U.S. Census Bureau. 2026c. "Business Formation Statistics." Public data files.

U.S. Census Bureau. 2026d. "Monthly State Retail Sales." Experimental public data files.

Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. "Where is the Land of Opportunity?" *Quarterly Journal of Economics* 129(4): 1553-1623.

Chodorow-Reich, Gabriel. 2020. "Regional Data in Macroeconomics." In *Handbook of Regional and Urban Economics.*

Davis, Steven J., and John Haltiwanger. 1992. "Gross Job Creation, Gross Job Destruction, and Employment Reallocation." *Quarterly Journal of Economics* 107(3): 819-863.

Decker, Ryan, John Haltiwanger, Ron Jarmin, and Javier Miranda. 2014. "The Role of Entrepreneurship in US Job Creation and Economic Dynamism." *Journal of Economic Perspectives* 28(3): 3-24.

Einav, Liran, Chiara Farronato, and Jonathan Levin. 2016. "Peer-to-Peer Markets." *Annual Review of Economics* 8: 615-635.

Fairlie, Robert W. 2020. "The Impact of COVID-19 on Small Business Owners." NBER Working Paper No. 27309.

Haltiwanger, John, Ron Jarmin, and Javier Miranda. 2013. "Who Creates Jobs? Small versus Large versus Young." *Review of Economics and Statistics* 95(2): 347-361.

Jordà, Òscar. 2005. "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review* 95(1): 161-182.

Klein, Philip A., and Geoffrey H. Moore. 1983. "The Leading Indicator Approach to Economic Forecasting." *Journal of Forecasting* 2(2): 119-135.

Stock, James H., and Mark W. Watson. 2002. "Macroeconomic Forecasting Using Diffusion Indexes." *Journal of Business and Economic Statistics* 20(2): 147-162.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225(2): 175-199.